

Exercise 3: Stored Procedures

Tables:

Customers

	CUSTOMERID	NAME	DOB	BALANCE	LASTMODIFIED	ISVIP
1	1	John Doe	5/15/1955, 12:00:00	15000	6/23/2026, 12:17:20	YES
2	2	Jane Smith	7/20/1990, 12:00:00	5000	6/23/2026, 12:17:20	(null)

Accounts

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE	LASTMODIFIED
1	1	1	Savings	1000	6/23/2026, 12:17:20
2	2	2	Checking	1500	6/23/2026, 12:17:20

Loans

	LOANID	CUSTOMERID	LOANAMOUNT	INTERESTRATE	STARTDATE	ENDDATE
1	1	1	5000	4	6/23/2026, 12:17:20	7/13/2026, 12:17:20
2	2	2	7000	6	6/23/2026, 12:17:20	7/18/2026, 12:17:20

Scenario 1: The bank needs to process monthly interest for all savings accounts.

Question: Write a stored procedure ProcessMonthlyInterest that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
IS
BEGIN
    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'Savings';

    COMMIT;
END;
/

BEGIN
    ProcessMonthlyInterest;
END;
/
```

Output:

1	SELECT	AccountID,
2		AccountType,
3		Balance
4	FROM	Accounts;

Query result		Script output	DBMS output	Explain Plan	SQL
 		Download	Execution time: 0.005 seconds		
	ACCOUNTID	ACCOUNTTYPE	BALANCE		
1	1	Savings	1010		
2	2	Checking	1500		

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

Question: Write a stored procedure UpdateEmployeeBonus that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (  
    dept_name VARCHAR2,  
    bonus_percent NUMBER  
)  
IS  
BEGIN  
    UPDATE Employees  
    SET Salary = Salary + (Salary * bonus_percent / 100)  
    WHERE Department = dept_name;  
  
    COMMIT;  
END;  
/  
  
BEGIN  
    UpdateEmployeeBonus ('IT',10);  
END;
```

Output:

1	SELECT EmployeeID,
2	Name,
3	Department,
4	Salary
5	FROM Employees;

Query result

Script output

DBMS output

Explain Plan

SQL history

 

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Execution time: 0.009 seconds

	EMPLOYEEID	NAME	DEPARTMENT	SALARY
1	1	Alice Johnson	HR	70000
2	2	Bob Brown	IT	66000

Scenario 3: Customers should be able to transfer funds between their accounts.

Question: Write a stored procedure TransferFunds that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

```
CREATE OR REPLACE PROCEDURE TransferFunds(  
    source_acc NUMBER,  
    target_acc NUMBER,  
    transfer_amount NUMBER  
)  
IS  
    current_balance NUMBER;  
BEGIN  
  
    SELECT Balance  
    INTO current_balance  
    FROM Accounts  
    WHERE AccountID = source_acc;  
  
    IF current_balance >= transfer_amount THEN  
  
        UPDATE Accounts  
        SET Balance = Balance - transfer_amount  
        WHERE AccountID = source_acc;  
  
        UPDATE Accounts  
        SET Balance = Balance + transfer_amount  
        WHERE AccountID = target_acc;  
  
        COMMIT;  
  
    END IF;  
  
END;  
/  
  
BEGIN  
    TransferFunds(1,2,500);  
END;  
/
```

Output:

1	SELECT AccountID,
2	Balance
3	FROM Accounts;

Query result Script output DBMS output Explain P			
Download Execution time: 0.002 seconds			
	ACCOUNTID	BALANCE	
1	1	500	
2	2	2000	